

Automated Detection of Coffee Leaf Diseases Using Convolutional Neural Networks and Transfer Learning

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Abstract. Coffee leaf diseases are responsible for significant yield losses in producing regions and are traditionally diagnosed through visual inspection by agronomists, a process that is slow and dependent on expert availability. This paper investigates the use of convolutional neural networks and transfer learning to automatically identify three common coffee leaf conditions – rust, brown eye spot, and cercospora leaf spot – directly from smartphone images. A dataset of 2,450 leaf images was collected in commercial plantations in the Ponta Grossa region and expanded to 6,320 samples through data augmentation. Four pretrained architectures (MobileNetV2, ResNet50, VGG16, and EfficientNetB0) were fine-tuned on this dataset and compared in terms of classification accuracy and inference time. The EfficientNetB0 model achieved the best result, with an accuracy of 96.4%, outperforming the other architectures while requiring fewer computational resources. These results indicate that lightweight transfer-learning models are a viable option for on-device disease screening in precision agriculture applications.

Keywords: machine learning, convolutional neural networks, transfer learning, plant disease detection

1. Introduction

Coffee is one of the most economically important agricultural commodities worldwide, and losses caused by foliar diseases can reach a substantial fraction of a harvest when outbreaks are not identified early (ANTONELLI; SMITH; SIMMONDS, 2019). Diagnosis is traditionally carried out by agronomists through visual inspection of the leaves, a process that is time-consuming, dependent on expert availability, and susceptible to inconsistency between evaluators (COPE *et al.*, 2012).

Computer vision and deep learning have been increasingly explored as a way to automate plant disease and species identification, allowing non-experts to obtain a diagnosis directly from a smartphone photograph. The contribution of this paper is twofold: a) a new image dataset of coffee leaves affected by three common diseases, collected under field conditions; b) a comparative evaluation of four convolutional neural network architectures, using transfer learning, for the task of automatic disease classification.

2. Background

Convolutional neural networks (CNNs) are hierarchical, multi-layer feature extractors that have become the standard approach for image classification tasks, including plant and leaf identification (LEE *et al.*, 2015). When the number of labeled samples available for a given task is limited, transfer learning is commonly used: a model pretrained on a large, general-purpose dataset such as ImageNet is fine-tuned on the smaller, task-specific dataset, allowing the hierarchical features learned previously to be reused instead of training a network from scratch (ZHUANG *et al.*, 2020). This strategy has been shown to reduce both training time and the amount of labeled data required to reach a competitive accuracy.

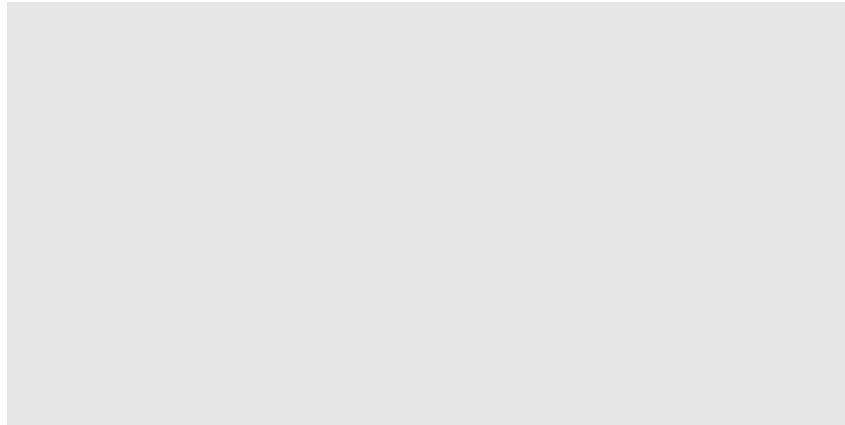


Figure 1 – Overview of the transfer-learning pipeline: a pretrained convolutional base extracts features from the leaf image, and a new classification head is trained on the coffee leaf disease dataset (adapted from (GARCÍA-ORDÁS *et al.*, 2020))

3. Methods

Leaf images were collected using smartphone cameras in commercial coffee plantations located in the Ponta Grossa region, Paraná, Brazil, between March and July 2024. Each image was labeled by an agronomist into one of four classes: healthy, rust, brown eye spot, or cercospora leaf spot. The initial set of 2,450 images was expanded to 6,320 samples using data augmentation techniques, including random rotation, horizontal flipping, and brightness adjustment, in order to reduce class imbalance and improve model generalization.

Four convolutional architectures pretrained on ImageNet – MobileNetV2, ResNet50, VGG16, and EfficientNetB0 – were used as feature extractors. For each architecture, the original classification head was replaced by a new fully connected layer trained on the coffee leaf dataset, while the convolutional base was kept frozen during the first training stage and partially unfrozen for fine-tuning in a second stage. A total of 80% of the images were used for training and 20% for validation, with 60 training epochs for each model.

4. Results

Table 1 summarizes the classification accuracy obtained for each architecture on the validation set, together with the average inference time per image measured on a mid-range mobile GPU. EfficientNetB0 achieved the highest accuracy (96.4%) while also presenting the lowest inference time among the four models, making it the most suitable candidate for deployment in a mobile screening application. ResNet50 obtained a comparable accuracy (95.1%) but at a substantially higher inference cost, whereas MobileNetV2 offered the best trade-off between accuracy and computational cost among the remaining models.

Table 1. Validation accuracy and average inference time per image for each model.

Metric / Model	MobileNetV2	ResNet50	VGG16	EfficientNetB0
Accuracy	0.93	0.95	0.89	0.96
Inference time (ms)	42	118	96	38

5. Conclusion

This paper presented a new coffee leaf disease image dataset and a comparative evaluation of four convolutional neural network architectures for automatic disease classification using transfer learning. EfficientNetB0 obtained the best overall result, combining the highest accuracy with the lowest inference time, which supports its use in mobile, on-device screening applications for precision agriculture. Future work includes expanding the dataset to additional diseases and crop varieties, as well as evaluating the models under field conditions with varying lighting and occlusion levels.

References

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